

A Preliminary Usability Evaluation of an Artificial Intelligence-Based, Motion-Detecting Wearable Device: The Geriatric Functional Assessment System

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Abstract

Background: Physical function is a key determinant of independence among older adults. Yet, there are barriers to assessing physical function in clinic. We developed a wearable geriatric functional assessment system (GFAS) that quickly and effortlessly evaluates physical function.

Methods: A single-arm, nonrandomized, mixed-methods, usability study evaluated the design, ergonomics, and usability of the GFAS. Participants aged >65 years with multiple chronic conditions were recruited and wore the GFAS about the clinic for 15 min. We assessed walking speed, 30-s sit-to-stand, and evaluated postural balance using a Footscan pressure plate system. In addition to transcribed exit-interviews and a Willingness to Pay questionnaire, the USE, Technology Acceptance, and System Usability Scales were evaluated.

Results: Of the 37 participants screened, 21 were recruited, enrolled, and consented, whose mean age was 76.6 ± 5.45 years (52.4% female), with 28.6% non-White, 19% were on Medicaid, and 52.4% were classified as having a robust Fried frailty status. Participants favored the prototype and its existing functionality (7.14 ± 2.35) and would wear it if recommended by their clinician (7.62 ± 2.50 , median 8.0). All felt it was easy to use, 74% of comments outlined they would use it again, and 81% noted it was comfortable. System Usability score was 78.2 ± 14.5 , USE was 5.83 ± 3.59 , and Technology Acceptance demonstrated satisfaction of 7.05 ± 2.18 in using the device.

Conclusions: The GFAS prototype shows considerable promise in evaluating physical function in older adults and that additional steps are needed to maximize usability.

Keywords: Physical function, Geriatric functional assessment system, Usability

Optimizing physical function is key for maintaining independence in older adults. Aging arises the onset of chronic medical conditions (1), changes in body composition (2), and an increased risk of developing frailty, which leads to a risk of disability, morbidity, and death (3). As interactions with the health system rise resulting from multimorbidity, the ambulatory setting provides a unique opportunity for healthcare practitioners to capture objective information on an individual's current physical function. Importantly, in primary care, long-term relationships and ongoing visits allow practitioners to also evaluate changes in individual function over time. Such changes can provide clinicians the necessary data to identify individuals who may benefit from targeted interventions to slow the decline in physical function while also providing the opportunity to implement care planning, all while optimizing an individual's ability to stay home independently.

Despite the strong and robust associations between physical function and important clinical outcomes such as morbidity,

disability, and institutionalization, objective performance measures, including gait speed, grip strength, and sit-to-stand, are rarely collected in clinical practice despite them being routinely performed in clinical trials. In contrast, busy primary care workflows prevent clinicians from routinely collecting such objective measures in clinical practice (4–9) despite their suitability for this setting. For instance, routine annual wellness visits designed for preventive care among older Medicare beneficiaries inquire about subjective physical function but do not mandate the objective collection of such measures. There continues to be an ongoing reliance of evaluating physical function based on subjective “visual” evaluations or by using survey-based measures. Neither of these modalities have sufficient diagnostic accuracy to identify individuals who may be at risk for functional decline, leading to a missed opportunities for prevention or treatment among older adults (10).

Wearable sensor-based devices may advance the field of healthy aging by enhancing the feasibility of collecting

objective assessments in busy primary care practices. While there have been significant advances in the design engineering field in terms of motion and visual capturing capabilities of devices, advances in machine learning techniques have enhanced the potential for transdisciplinary team science to translate these novel technologies from laboratory to clinic-based settings. However, many technological advances focus on one single physical function measure rather than integrating multiple measures. For instance, a patient may have normal balance but have impairments in strength or vice-versa. Capturing multimodal measures of physical function and creating a model to identify trajectories of function hold great potential for creating benchmarks to initiate treatment regimens in the future.

With funding from the Johns Hopkins Artificial Intelligence Technology Center, our team conducted an early-stage usability evaluation of a geriatric functional assessment system (GFAS) in older adults that could be applied in an ambulatory clinical setting. Our purpose was not only to design this device and create novel machine learning algorithms to infer physical function based on objective performance measures but also to test the human factors responsible for future deployment in clinical practice. We envisioned that this pilot study would provide foundational data to expand this line of research in the hopes of incorporating this into clinical care.

Method

Study Setting/Design

We conducted a single-arm, nonrandomized, prospective, mixed-methods, usability study to evaluate GFAS in a primary care clinical setting between January 18th and March 7th, 2024. The purpose of this single-visit study was to provide insights into its design and ergonomics and to collect data to develop the necessary machine learning algorithms to infer physical function. The University of North Carolina (UNC) at Chapel Hill's Human Subject Research Project Program approved the study.

Recruitment/Sampling

A convenience sample was recruited from the Geriatrics Clinic at the UNC multispecialty clinic building located in Chapel Hill, NC. This clinic serves over 3 600 patients with a mean age of 80. We posted recruitment advertisements on institutional listservs and the North Carolina Translational Science Institute Research for Me website, and recruited patients attending Geriatrics Clinic visits. Clinic schedules were screened by the research assistant, and after medical record review, participants were approached to see whether they had interest in the study. All participants provided informed consent. Eligible participants were aged over 65 years with at least two chronic medical conditions (alcohol abuse; Alzheimer's disease and related dementia; heart failure; arthritis [osteoarthritis or rheumatoid]; chronic hepatitis; asthma; HIV/AIDS; atrial fibrillation; hyperlipidemia; autism spectrum disorder; hypertension; cancer; ischemic heart disease; chronic kidney disease; osteoporosis; chronic obstructive pulmonary disease; schizophrenia or other psychotic disorders; depression; stroke; or diabetes). We excluded participants that were not patients from our institution. A Calhoun Cognitive Screening questionnaire was conducted to ascertain capacity for consent (11). A paper consent form was given to participants prior to their visit date. The study team

described the study procedures in detail, and if agreeable, the participant completed the consent form at the study visit.

Study Procedures

Study visits occurred at the UNC multispecialty clinic building. All participants received a brief, interactive session on how to use the device. The device was then placed on participants (see details below) and the device was turned "on," while the participant walked around the clinic environment for 15 min. At its conclusion, the research assistant performed physical function assessments and an exit interview (Supplementary Material 1). Participants were subsequently asked to complete surveys as outlined below. The survey instruments were completed in-clinic. All participants were provided a \$25 gift card.

Device Prototype

The objective of designing the prototype was to assess and gain information on physical function through motion- and visual-based user data (Figure 1). Motion data collection was facilitated by an Arduino Inertial Measurement Unit (IMU) shield mounted on an Arduino MKR Zero board, which was integrated with a micro-SD card to store the captured data. The IMU shield recorded four types of motion data—accelerometer, gyroscope, magnetic field, and Euler angles—allowing for a detailed analysis of user movements and orientation. For visual data, an Arduino Portenta H7 board was paired with an Arduino Vision Shield, which featured a high-resolution camera module. The Portenta H7 processed and stored images on a micro-SD card, ensuring accurate and readily accessible visual data. The use of Arduino-based hardware provided a flexible and customizable platform, enhancing the device's robustness and versatility. To support prolonged use, the device was powered by a 10 000 mAh battery.

To house these components, the encasing was designed using 3D printing and underwent 23 iterations to optimize functionality and usability. Since the device is worn around the neck, achieving proper weight distribution was a key consideration. To ensure balance and minimize discomfort, the chips

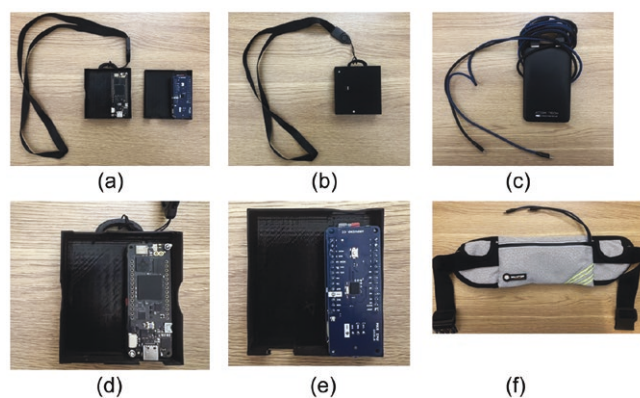


Figure 1. Geriatric functional assessment system. Images of the device in various forms, two including the exposed inner pieces of the device. One half of the device contains the microchip that collects the motion data, while the other half contains the microchip that collects the image data. The two images on the right show the battery pack used to charge the device and how it is presented to the participant. (A) Device prototype—opened; (B) device prototype—closed; (C) battery along with wires; (D) vision module inside the device; (E) motion sensor inside the device; and (F) pouch used to keep battery and wires.

were strategically placed on opposite sides of the encasing. The design also incorporated practical features, including slots for battery connections, micro-SD card exchange, and LED indicators for various operations. A hook at the top allowed for neckband attachment, while curved side features facilitated easy access to internal components.

Objective Measures

In addition to conducting vital signs (height, weight, blood pressure, oxygen saturation, and respiratory rate), the research assistant conducted three key physical function assessments with participants: (i) Walking speed: In an 8-m course, we measured the time to walk 4 m with a participant's usual walking speed and stride was measured. Participants started walking for the first 2 m, at which time the RA started the stopwatch and stopped timing ended at the 6-m walk—there was a 2-m walking phase that was not timed at the end. This provided a run-in and run-out period, and the mean of three trials was recorded; (ii) 30-s sit-to-stand: participants sat in a chair with arms folded. The number of repetitions of standing up and returning to sit over 30 s was recorded. A repetition was counted once the participant's bottom touched the seat of the chair; (iii) Postural Balance: A Footscan Entry Level V9 1-meter Pressure Plate System (Leuven, Belgium) measured each participant's center of pressure in millimeters (mm) during a double-leg stance. The Center of Pressure was recorded along both the X (side-to-side) and Y (front-to-back) directions. Additionally, we calculated the total distance traveled by the Center of Pressure in mm and the ellipse area (EA) in square millimeters (mm²), which represents the area enclosing the total Center of Pressure trajectory. Data were collected for 30 s, with a start delay of 5 s per trial, and three trials were performed. Participants stood barefoot on the platform with feet 5 cm apart at the heels and with the toes angled out forming about a 30° angle. Participants looked straight ahead at an image of the correct foot placement posted on the wall. Data were stored in a HIPAA-compliant database.

Frailty Status

We evaluated each individual's phenotypic frailty status using Fried's criteria (12) based on a mixture of subjective and objective measures, aligning with our previous methods (3). Criteria included: (a) gait speed: <0.8 m/s; (b) low activity: based on a physical activity compared with peers their age; (c) exhaustion: whether participants felt tired during the day; (d) weakness: measured if tired over the past month; and (e) weight loss of >10% in the past 6 months—as not all participants had weight at 6 mo, we used the weight closest to the 6-mo date. Participants were categorized as frail if they fulfilled three criteria, prefrail with one or two criteria, and robust if no criteria were fulfilled.

Questionnaires

Participants were asked how often they would come to clinic for evaluation and enquired by posing single Likert questions on the device's overall acceptability and preference (Likert 1–10, low to high). To assess the system's usability, participants completed three self-reported questionnaires on the use of the device: (i) The USE questionnaire (13,14) is a validated tool that evaluates the user experience across domains of usefulness, satisfaction, ease of use, and ease of learning of using a system. This questionnaire was adapted to exclude items

that were not applicable to this specific technology-based system; (ii) Technology Acceptance (15,16) is a questionnaire that helps evaluate the acceptability of a given system to participants and whether they would be willing to use it; and (iii) The System Usability Scale (17,18) is often used on small samples for technology usability (range 0–100) to evaluate a system's ease of use. A one-item Willingness to Pay (19) question determined that the relative value participants would place on the device in return for information on their health. Likert scales (1–10, low to high), were also used to evaluate privacy concerns and concerns regarding integration into the electronic health record (EHR). All questionnaires were completed through REDCap, an electronic data entry capture system.

Exit Interviews

All interviews were conducted by the research assistant and audio-recorded. They were transcribed by a professional transcription service and uploaded into a qualitative data analysis program, *Dedoose*, for coding. Questions focused on device acceptability, barriers to use, ergonomics (eg, comfort), and interactions with the study personnel (Supplementary Material 1). Interviews lasted ~10 min consisting of semi-structured questions, clarifying probes in which we encouraged participants to elaborate upon open-ended questions. All recordings were de-identified upon transcription and aggregated, stored on password-protected, institutional servers.

Statistical Analysis

All analyses were conducted in both Python and in Microsoft Excel 365 (Redmond, WA). Data were exported from REDCap and aggregated into a single dataset. Descriptive characteristics as mean $\pm$ SD and count (percentage) and median and range are presented where appropriate. The primary outcomes of the study were measures of usability (USE questionnaire, System Usability, Technology Acceptance). For data analysis, three machine learning models evaluated the accuracy of each in inferring changes in physical function status (improvement, stability, or decline). All interview data were transcribed, coded, and summarized.

Machine Learning Analysis

We employed support vector classifiers (SVCs), random forests (RFs), and deep neural networks (DNNs) for motion-based classification tasks. Support vector classifiers were selected for their ability to handle high-dimensional data and identify optimal decision boundaries, particularly in nonlinearly separable cases. Random forests were included for their capacity to model nonlinear relationships and provide feature importance insights, while DNNs were chosen for their effectiveness in learning complex hierarchical patterns from large, variable datasets. Initial comparative testing against simpler models, such as k-nearest neighbors and logistic regression, confirmed the superior performance of these selected methods.

The models were trained using features extracted from accelerometer, gyroscope, Euler angle, and magnetic field data, incorporating statistical measures such as mean, variance, skewness, and kurtosis across the XYZ axes. Key clinical outcomes assessed included gait speed, sit-to-stand transitions, and balance. Data processing involved computing mean values within predefined windows to capture relevant movement patterns, classifying physical function into three categories: improved, stable, or declined. Improvement

was characterized by increased gait speed, more sit-to-stand repetitions, and reduced travel distances, while a decline reflected slower gait, fewer transitions, and increased travel distances.

For robust model evaluation, the dataset was split into 70% training, 15% validation, and 15% testing, ensuring a balanced assessment of performance. Due to the structured data pairing approach, additional cross-validation was unnecessary, as it effectively expanded the dataset and minimized reliance on data augmentation techniques. This strategy enhanced variability across subsets, strengthening the models' robustness without requiring additional sampling methods.

Results

Of the 37 screened participants, 21 were enrolled. Of the 16 participants who did not enroll, all fulfilled our selection criteria. Despite expressing interest, three did not return our phone calls, two declined to participate in research activities, and the remaining did not want to engage with technology. Baseline characteristics are outlined in Table 1 and Supplementary Material 2. Mean age was 76.6 ± 5.5 years. Approximately half (52.4%) were female, 28.6% were non-White, 19% were on Medicaid, and 47.6% had a household income $< \$50,000$. Approximately 52.4% were classified as robust, with a mean walking speed of 1.13 ± 0.27 m/s and 30-s sit-to-stand repetitions of 11.38 ± 4.34 (median 11). Overall, participants favored the prototype and its existing functionality (7.14 ± 2.35 , range, low to high, 1–10). Participants were willing to use this device if they were to attend clinic every few months ($n = 14$ [66.6%]) to assess their physical function. Individuals did not believe that they would need more practice to use the device (4.29 ± 3.57 , 1–10 scale, no need to high need) and would feel comfortable using this device, even at home, if recommended by their clinician (7.62 ± 2.50 , median 8.0, 1–10 scale, low to high comfort).

If my physician told me they were concerned about something, I would totally [use the device]. No problem at all.

I was thinking it would be monitoring me and seeing what my organic age as opposed to my numerical age. That's the word for it. What age is my body developed at this stage is what you could see how you're doing.

I think the information would be helpful to monitor progress or no progress or going backward instead of forward.

Participants had mixed feelings as to whether they would consider wearing such a device continuously outside of the clinic (5.8 ± 2.89 , 1–10 scale, low desire to high). Overall, participants expressed the need for more information to understand its use and noted that additional design engineering features were needed to make it more aesthetically pleasing.

Well, it's bulky. It would be nice if it was a little less bulky and if the device was like the size of a medallion, a locket or something.

Except for that little box, I think that could be tweaked a bit to be more less intrusive maybe or more I was just really aware of it.

It seemed pretty crude, this black box, this heavy power pack on your waist. Seemed like it was very much a prototype.

Table 1. Baseline Characteristics

Characteristics	Value
Age, years	76.6 ± 5.45
Female sex	11 (52.4%)
Race	
White	15 (71.4%)
Black	6 (28.6%)
Health insurance	
Medicare	20 (95.2%)
Medicaid	4 (19.0%)
Private	1 (4.8%)
Smoking status	
Nonsmoker	11 (55.0%)
Former	8 (40.0%)
Current	1 (5.0%)
Education	
High school	5 (23.8%)
Some college	2 (9.5%)
College	6 (28.6%)
Postcollege	8 (38.1%)
Household income	
$< \$25K$	5 (23.8%)
\$2549 999	5 (23.8%)
\$5074 999	3 (14.3%)
\$75 000+	8 (38.1%)
Comorbidities	
Alcohol abuse	1 (4.8%)
Alzheimer's disease and related dementias	1 (4.8%)
Arthritis	9 (42.9%)
Asthma	3 (14.3%)
Cancer	4 (19%)
Chronic kidney disease	6 (28.0%)
COPD	3 (14.3%)
Depression	7 (33.3%)
Diabetes	6 (28.6%)
Hepatitis	1 (4.8%)
HIV/AIDS	1 (4.8%)
Hyperlipidemia	14 (66.7%)
Hypertension	15 (74.4%)
Osteoporosis	2 (9.5%)
Other psychotic disorders	1 (4.8%)
Frailty status	
Robust	11 (52.4%)
Prefrail	8 (38.1%)
Frail	2 (9.5%)

Notes: All values represented as mean $\pm$ SD or count (%). COPD = chronic obstructive pulmonary disease.

Importantly, all participants felt it was easy to use, 74% of comments outlined they would use it again, and 81% noted it was comfortable.

You just plug it in, put it on, and not doing anything else, except maybe being careful with the camera.

It's easy to use. I would not need an assistance to use it and I am assuming it serves its purpose

It seems to be easy to use. Probably needs a better battery. It doesn't last very long. If it provides what doctors are looking for about us, I think it's great device. I think it's a great idea. Once you get it working, I think it'll be a great device to record stuff, and it would be great if it was wireless so they wouldn't have to get back. They could get the information automatically from it.

We noted variation in the USE questionnaire (Usefulness, Satisfaction, and Ease of Use). While the overall score was marginally above average at 5.83 ± 3.59 (1–10, low to high), we found higher scores in ease of use and learning and lower scores in usefulness. Technology acceptance demonstrated different results in satisfaction, with favorable results in confidence, usefulness, efficiency, and overall attitudes (Table 2, Supplementary Material 3 and 4).

Technology helps you to understand that the body, while it's maturing, it has to get improvements so you won't get sick. With technology, it helps a lot of the older people.

I think that because I'm old, I'm kind of like if you show me and you write it -- if I have repetition with it, then I'm okay. If it seems like it would have a good purpose to me, then that would help me.

Make it more useful for me

System Usability Score, which is critically important in the design engineering features of newly designed systems, was extremely promising at a 78.2 ± 14.5 , exceeding the industry standard of 68 (20).

I know from my own personal self, if I had feedback every day, that would help me assess my own situation.

I thought it was a good way to measure the... it was walking skills

The majority of participants (52.3%) would not be willing to pay more than \$25 for this device at home. Individuals reported low concern about privacy (4.24 ± 3.55 , median 3, 1–10, low to high), and low concern about integration with the EHR (4.38 ± 3.44 , median 3, 1–10, low to high).

Table 2. Summary Measures of Usability Questionnaires

	Domain	Mean $\pm$ SD
USE questionnaire*	Overall score	5.83 ± 3.59
	Usefulness	4.43 ± 3.62
	Ease of use	6.59 ± 3.25
	Ease of learning	7.29 ± 3.48
	Satisfaction	5.81 ± 3.44
Technology acceptance*	Confidence	7.95 ± 2.77
	Overall attitudes	6.95 ± 2.42
	Bothersome	4.40 ± 2.84
	usefulness	7.05 ± 2.46
	Satisfaction	7.05 ± 2.18
	Efficiency	7.02 ± 2.06
System usability score#	Overall Score	78.2 ± 14.5

Notes:

*Likert scores for questionnaires range from low to high (1–10).

#scores range from 0 to 100.

Design engineering evaluation

During testing, a flipping issue was observed in one instance, resulting in the loss of some image data. To address this, further refinements will be made to the encasing's dynamics to reduce the likelihood of flipping. Additionally, transitioning to an internal lithium polymer (LiPo) battery will eliminate the need for an external power supply, allowing for a more compact and lightweight design. By integrating the battery, chips, and other components into a single encased unit, these modifications can enhance the device's portability and usability, making it more suitable for long-term real-world applications.

Machine Learning Results

Model performance was assessed across three classification tasks—gait-speed prediction, sit-to-stand classification, and balance assessment. For gait-speed prediction, RF performed the best with an accuracy of 0.775, weighted precision of 0.842, weighted recall of 0.775, and weighted F1-score of 0.794. Neural network followed with an accuracy of 0.713, weighted precision of 0.719, weighted recall of 0.713, and weighted F1-score of 0.698. Support vector classifier had the lowest performance with an accuracy of 0.677, weighted precision of 0.634, weighted recall of 0.677, and weighted F1-score of 0.654.

For sit-to-stand classification, RF had the lowest accuracy at 0.565, with a weighted precision of 0.708, weighted recall of 0.565, and a weighted F1-score of 0.605. Neural network outperformed both models with an accuracy of 0.661, weighted precision of 0.611, weighted recall of 0.661, and weighted F1-score of 0.633. Support vector classifier had intermediate performance, achieving an accuracy of 0.654, weighted precision of 0.594, weighted recall of 0.654, and weighted F1-score of 0.621.

For balance assessment, SVC had the highest accuracy at 0.663, with a weighted precision of 0.683, a weighted recall of 0.663, and a weighted F1-score of 0.648. Neural network closely followed with an accuracy of 0.646, weighted precision of 0.666, weighted recall of 0.646, and weighted F1-score of 0.636. Random forest had the lowest accuracy at 0.585, with a weighted precision of 0.701, a weighted recall of 0.584, and a weighted F1-score of 0.624.

Overall, RF performed best for gait-speed prediction, NN excelled in sit-to-stand classification, and SVC was the top performer for balance assessment.

Discussion

Our usability study demonstrates the feasibility of deploying a GFAS in collecting objective physical function data while concurrently generating a positive user experience among an older adult population within an ambulatory clinic setting. These findings support the further development and optimization of this device to improve its ergonomics, validity, and demonstrate its efficacy.

As with any new technology-based design, iterative design engineering principles need to be applied to ensure that the end-user is able and willing to use the system. We focused on the participant end-user, recognizing that we would require additional steps to improve the device's functionality and usability and that additional design features would be required for provider-facing components. While older adults

are the fastest growing segment using technology, they continue to have visual and auditory barriers in using technology-based devices. One major advantage of the GFAS is that it does not considerably rely on the user. This is unlike other devices, including mobile phones, or other wearables which make comfort, ergonomics, and aesthetics much more important, than its ease of use.

A major concern that was raised by the participants was that they were unclear how such a device could potentially help them in the future. Understandably, commercial devices are often targeted toward consumers rather than healthcare providers or patients within healthcare systems (and being paid for by third party insurers). Hence, we acknowledge that this may have affected our results in a negative manner and future studies should consider further explanations by our research staff to participants. A component that was not evaluated in this specific study was the clinician end-user—that is, what type of data output and dashboard could be useful for clinicians in interpreting these findings and implementing geriatric-specific care plans. Such engagement will harmonize and balance opinions of different groups in future design iterations.

As an early-stage prototype, our usability findings were promising and exceeding our expectations. Specifically, a System Usability Score of 68 often is a cutoff for moving a system along its development to the next stage—our findings exceeded this number. However, we are cautious by these promising results. While participants expressed quantitatively their impressions, our qualitative findings and evaluations on other questionnaires tempered these expectations. Our study also uses rigorous methods including mixed-methods to integrate results in informing our technology development. This is a vital step in any systems-engineering approach for designers and in considering future deployment. Importantly, our pilot had adequate representation from non-White participants with low income on Medicaid, representative of lower socioeconomic status. We also ascertained frailty status, which approached representation of a usual community-based cohort (21), suggesting that our results have external validity.

As with any early-stage usability evaluation, our study is not without limitations. Our sample of 21 participants demonstrated our ease of recruitment and in conducting the study measures—there was high acceptability among the study logistics and procedures. However, while our system's accuracy was promising, a larger test dataset will be needed to create validated algorithms. Yet, we achieved our target recruitment and were able to isolate conceptual themes within the ranges for theoretical saturation. Additionally, our “ground-truth” was established using three measures of physical function—we acknowledge that there are other measures of physical function that assess different domains (eg, performance, strength, balance) that may better characterize function, and that additional data, including self-report, caregiver, or other vital signs may be required. Furthermore, we have yet to deploy the device over a long period of time which may impact usability and data collection within clinical settings. Finally, as with any usability evaluation, there may be barriers or facilitators unique to the population examined that may have not been revealed yet.

In addition to the technical adaptations to the machine learning algorithms, our results suggest potential adaptations to the device. First, from a design perspective, the device needs

to be smaller, more compact, and lighter. Second, the battery and device need to be more efficient. Third, the strap holding the GFAS was very uncomfortable to most participants, which can easily be changed. Fourth, we need to incorporate additional measures of physical function—including grip strength, timed up and go, tandem balance, and 6-min walk. These may include subjective functional measures as well that incorporate patient-reported outcomes.

Key usability measures of the GFAS demonstrated promise that could direct our future research efforts. In addition to increasing the sample size and continue iterative, user-centered design testing among older adults, longitudinal testing, and monitoring is necessary to demonstrate changes in output measures over time and ensure that participants continue to feel comfortable in using this device in clinical settings. Once the design engineering features, including battery life, form factor, and other electronic-based components are finalized, data pipelines will need to be created to transmit data either to EHR or cloud-based dashboards. Our design team should take each of these into account and conduct further user testing in future iterations.

The development of a prototype device exemplifies the integration of transdisciplinary team science, effectively bridging critical clinical needs with advanced computational methodologies. Within our healthcare system, there is an urgent need to enhance operational efficiency, and the GFAS holds significant potential to address these challenges by transitioning from subjective assessments of physical function to objective, longitudinal measures capable of tracking individual trajectories over time. Although the integration of EHR-based algorithms is in its early stages, such advancements are likely to streamline clinical workflows and enhance decision-making processes. Access to longitudinal data is essential for identifying critical inflection points in a patient's health trajectory, enabling algorithm-driven prompts that facilitate proactive, rather than reactive, clinical interventions. Future research, particularly through qualitative methodologies, is required to elucidate the actionable steps necessary for optimizing the clinical utility and implementation of such systems.

When implementing technology-based devices, particularly those used for collecting sensitive health data in at-risk populations, it is essential to address ethical considerations related to privacy, security, and data protection. Safeguarding participant information requires adherence to the highest ethical and regulatory standards, ensuring compliance with contemporary guidelines such as the Health Insurance Portability and Accountability Act (HIPAA) and other relevant data protection frameworks. Future iterations of the system must incorporate robust security measures, including local encryption behind firewalls, with data uploaded exclusively to secure, institutionally approved local servers. To further enhance privacy protections, algorithmic processing should ideally occur within the local system rather than relying on cloud-based or external computing resources. Additionally, there should be a clear and intentional separation between HIPAA-protected identifiers and other collected data to minimize the risk of unauthorized access or breaches. As the system is developed, these security and privacy protocols must be explicitly defined, documented, and iteratively refined to align with emerging best practices and technological advancements. Collaboration with cybersecurity experts, institutional review boards, and data governance committees will be

essential to ensuring that the system remains compliant, secure, and ethically sound.

Ultimately, our system will provide an effective means of measuring objective physical function in older adults attending ambulatory clinics in a validated manner, permitting documentation within the EHR, and determining benchmarks and categories of change. If successful, incorporating such wearable monitoring within clinical practice may encourage Medicare or third-party payors to incentivize providers to provide reimbursement when such measures are recorded over time and incorporated into patient care plans and promoting evidence-based health promotion programs.

Supplementary Material

Supplementary data are available at *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences* online.

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Conflict of Interest

None.

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